PREP 4 · LESSONS 7–8

Power and wasted energy

Power as the rate of energy transfer · wasted energy · reducing it

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 State the law of conservation of energy. DN 2 [1]

Energy can be transferred usefully, stored or dissipated, but it cannot be created or destroyed.

2 Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis). DN 357 [1]

It is the gradient of the straight section.

3 Name the store of energy associated with extending and compressing objects. DN 12 [1]

The elastic potential energy store

Key ideas LESSONS 7–8

Show your working

4 A student climbs the stairs. They transfer 1500 J to their gravitational potential store in 6.0 s. Use $P = \frac{E}{t}$

Calculate the student's power. [3]

$P = \frac{E}{t}$ ✓

$P = \frac{1500}{6.0}$ ✓

$P = 250\,\text{W}$ ✓

5 A 2000 W kettle is switched on for 120 s. Calculate the energy transferred. [2]

$E = Pt$ ✓ (rearranged)

$E = 2000 \times 120 = 240\,000\,\text{J}$ ✓

6 Where does most wasted energy end up? [1]

the kinetic store of the device | **the thermal store of the surroundings** | the chemical store of the fuel

7 Explain how oiling a bicycle chain reduces wasted energy. [1]

It reduces friction, so less energy is transferred to the thermal store ✓

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